Quantifying comprehensive marine vessel emissions using data fusion

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Last Updated on July 10, 2025

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1 In-line manuscript statistics

- The transportation sector is a major source of global greenhouse gas (GHG) and local air pollution emissions, responsible for 21% of all CO_2 emissions globally in 2023
- For over 900,000 AIS-broadcasting vessels, we quantify point-estimate emissions for each vessel using precise latitude/longitude/timestamp information for over 97 billion individual AIS messages

2 Figures

2.1 Figure 1

2.1.1 Figure 1, panel A

2.1.2 Figure 1, panel B

2.1.3 Figure 1, combined

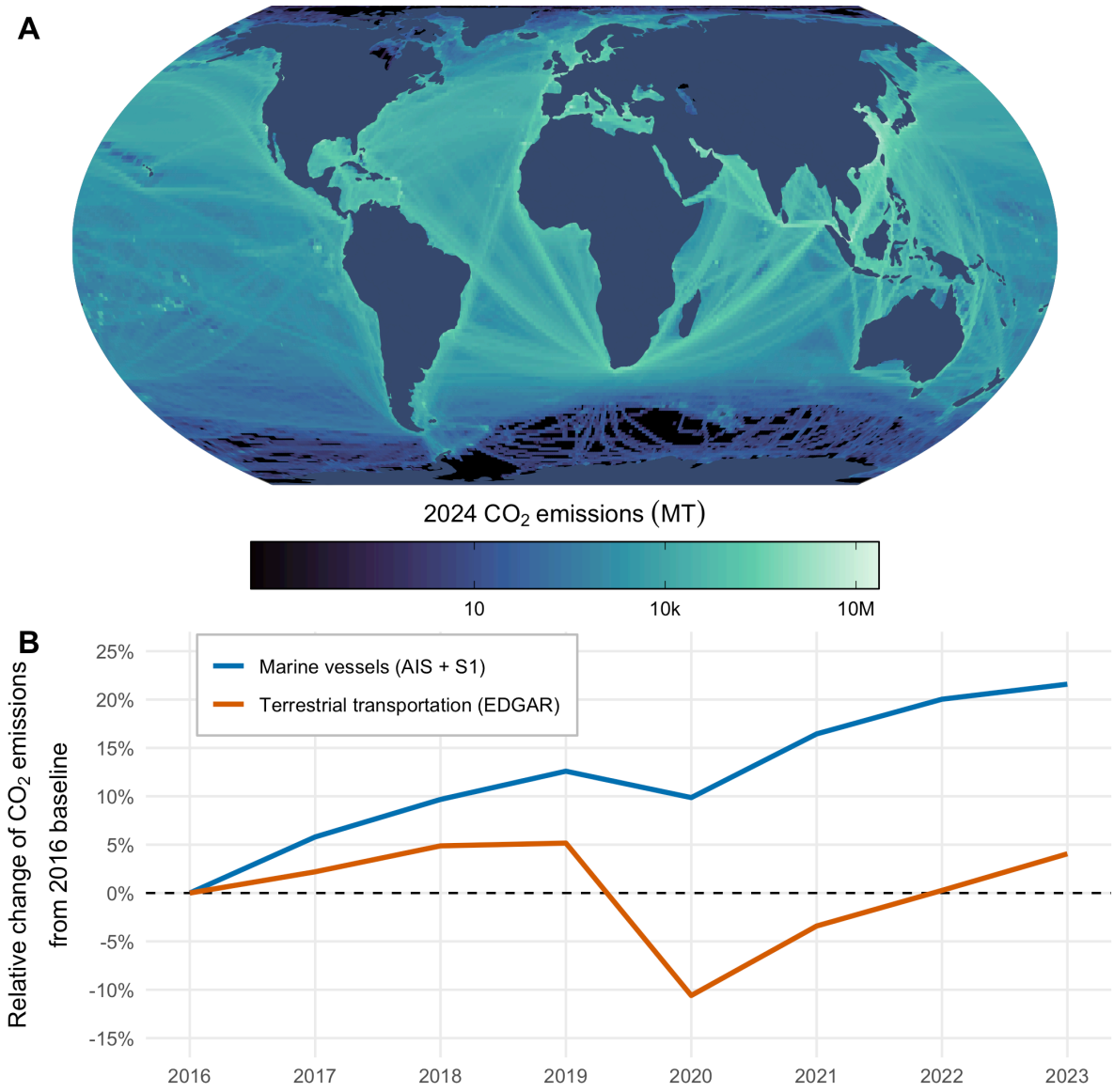

Figure 1

2.2 Figure 2

2.2.1 Figure 2, panel A

2.2.2 Figure 2, panel B

2.2.3 Figure 2 combined

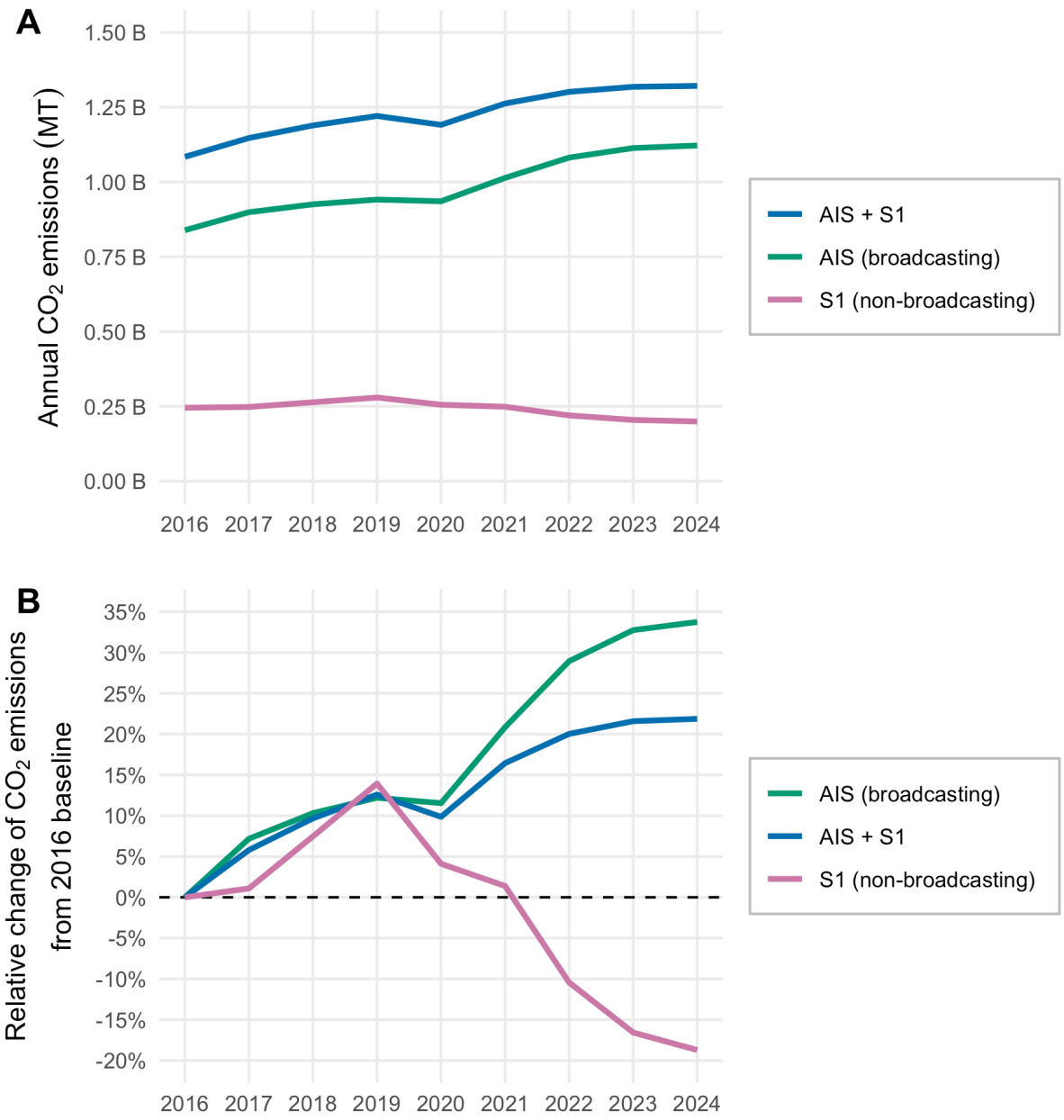

Figure 2: (A) Annual marine vessel CO2 emissions, with each line colored based on the data source. (B) Relative change in annual marine vessel CO2 emissions from a 2016 baseline, with each line colored based on the data source.